Sergei Kopylov

Research Engineer | Reasoning LLMs and Alignment

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Research engineer developing methods and infrastructure for reasoning LLM post-training and evaluation. Work includes verifiable rewards, rubric-guided LLM evaluation, and interpretability studies of attention and algorithmic reasoning.

Professional experience

Research Engineer (LLMs / Alignment) | Yandex — Nov 2025–present

- Fine-tune and align reasoning LLMs with supervised fine-tuning, RLHF, and RLVR; develop alignment methods and reasoning tasks for research experiments.
- Developed a Codeforces RLVR evaluation system with parallel solution execution and correctness verification, handling 1,000+ concurrent requests across simultaneous training runs.
- Supported RL training experiments reaching 61.8% pass@1 on an internal Codeforces C++ benchmark, versus a 57.9% baseline (+3.9 percentage points).
- Co-developed a Python execution environment for tool-integrated RLVR, enabling models to run code during reasoning; associated experiments improved performance on challenging real-world assistant tasks from 59.4% to 69.5% (+10.1 percentage points) and on AIME 2025 from about 86% to as high as 98%.
- Develop task-specific rubrics and LLM-as-a-judge evaluation methods for tasks in multiple domains without reliable programmatic reward signals.

Research Engineer Intern (LLMs / Alignment) | Yandex — Aug–Nov 2025

- Aligned reasoning LLMs with RLHF and RLVR; designed verification signals, integrated automated evaluators, and developed reasoning tasks for R&D experiments.
- Developed ranking-based RLVR tasks with deterministic, partial-credit rewards from human-ranked answers; evaluated mixed GRPO training. Research poster

Education

HSE University | Faculty of Computer Science — Sep 2023–Jun 2027

Bachelor's degree in Applied Mathematics and Information Science; GPA: 8.8/10.

Additional study: algorithms (Tinkoff Education), Python (Yandex Lyceum), reinforcement learning (YSDA), and NLP and deep learning projects at HSE. Completed all first-year courses in HSE's pilot cohort with top marks.

Selected projects

- **Human Preferences as Verifiable Rewards for GRPO (research poster):** Turned human answer rankings into automatically checkable rewards; mixed RLVR experiments improved selected response-quality metrics.
- **Codeforces RLVR Process-Credit Study:** Developed privileged value and sparse-distillation teachers for reasoning traces; found stronger outcome prediction with editorial context, but identified limits of value changes as automatic process-credit labels.
- **JPEG Decoder (C++):** Built a JPEG decoder with byte-level processing and RGB output.
- **Legal-document review with RAG:** Built a Python tool to check and revise contracts against a knowledge base of recommendations.

Research mentorship

Research Mentor | AI360 School — 2026

- Mentor seven students studying attention and information retrieval across LLMs, including Qwen3-8B and linear-attention models; guide Needle-in-a-Haystack experiments, controls, and ablations.

Research Mentor | Yandex School of Data Analysis (YSDA) — 2026

- Mentored two students studying Qwen3-8B reasoning on Codeforces 1660B using hidden-state and attention-head probes and controlled activation patching; distinguished predictive signals from causal steering.

Skills and achievements

Programming: Python, C++ **Research and ML:** LLM alignment, SFT, RLHF, RLVR, automated evaluation, transformers, RAG. **Interests:** NLP, bioinformatics.
Achievements: Winner of IOIP and Cognitive Technologies (2023); prizewinner in Technocup and St Petersburg State University informatics contests (2023), regional informatics (2022–2023), and regional mathematics (2021). Codeforces Expert (1714).